

# Expert Witness Letters — Final Templates

## Rambabu Vadlamudi — EB-1A Extraordinary Ability Petition

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### LETTER 1 — HEALTHCARE AI RESEARCHER / SENIOR LAB DIRECTOR (ACADEMIC OR INDUSTRY)

**Best for:** Faculty at UT Southwestern, Baylor Scott & White, Georgetown (JAMA 2025 PI), Google Cloud Healthcare, JAMA/XiFin researchers  
**Strongest EB-1A criteria served:** C5 (original contributions of major significance), national importance argument

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[DATE]

[LETTER WRITER NAME] [TITLE] [INSTITUTION / ORGANIZATION]  
[ADDRESS LINE 1] [CITY, STATE ZIP] [EMAIL] [PHONE]

U.S. Citizenship and Immigration Services National Benefits Center  
[Service Center Address]

#### **RE: Expert Letter in Support of EB-1A Extraordinary Ability Petition for Rambabu Vadlamudi**

Dear USCIS Officer:

I am writing in strong support of the EB-1A Extraordinary Ability petition filed on behalf of Rambabu Vadlamudi, Founder and Enterprise Architect of Ardia Health Labs in Argyle, Texas. My name is [NAME], and I serve as [TITLE] at [INSTITUTION]. My professional focus is [healthcare artificial intelligence / clinical informatics / molecular diagnostics / healthcare revenue cycle technology], and I have [X] years of experience in this domain. [Two to three sentences on credentials: academic positions, publications, leadership roles, peer recognition, e.g., “I have published more than [X] peer-reviewed articles on [topic], serve on the editorial board of [journal], and have received [award or recognition].”] I offer this letter based solely on my independent expert assessment of Mr. Vadlamudi’s technical contributions; I have no personal, financial, or employment relationship with him or with Ardia Health Labs.

#### **I. The National Problem Mr. Vadlamudi’s Work Addresses**

The United States independent clinical laboratory sector is in the midst of a reimbursement crisis that poses a direct threat to healthcare infrastructure. Independent laboratories — the 7,000-plus CLIA-certified facilities that operate outside hospital systems and provide approximately 32% of all US diagnostic test volume — face a structural disadvantage in medical claim reimbursement that no existing technology platform has adequately addressed.

The quantified scale of this crisis is striking. A 2024 analysis of more than 20 million laboratory claims by XiFin, Inc. documented a molecular diagnostics claim denial rate of 35.3% — the highest denial rate across all healthcare specialties, far exceeding radiology (approximately 14%), surgical specialties (approximately 11%), and primary care (approximately 7%). A 2025 Georgetown University study published in JAMA Network Open, analyzing 29,919 next-generation sequencing (NGS) claims, documented an overall NGS denial rate of 23.3%, rising to 27.4% following CMS National Coverage Determination (NCD) changes. Most critically, that study found, after multivariate adjustment for payer type, test complexity, and patient demographics, that independent laboratories face 2.76 times higher odds of claim denial compared to hospital-based laboratories. This disparity is not attributable to clinical quality differences; it reflects the structural resource asymmetry between independent laboratory billing operations and the payer AI systems evaluating their claims.

The economic consequence of this disparity is severe: despite documented appeal win rates of 50% to 80.7%, 65% of denied claims are never appealed, according to the Healthcare Financial Management Association and LigoLab's 2024 Laboratory Revenue Recovery Report. The gap between denial rates and appeal rates represents an estimated \$10 to \$12 billion in annual preventable revenue loss to the independent laboratory sector. The Protected Access to Medication Act (PAMA) reimbursement schedule compounds this pressure; CMS is projected to implement cumulative Medicare rate cuts of up to 45% by 2029, a prospect that led more than 50% of independent laboratory directors surveyed by the American Clinical Laboratory Association in 2024 to report considering consolidation or sale of their operations. Independent laboratories disproportionately serve rural communities, addiction medicine clinics, specialty oncology practices, and underserved urban populations — the communities that would be most harmed by laboratory consolidation.

## **II. Mr. Vadlamudi's Original Contribution — The DGP Architecture**

Against this backdrop, Mr. Vadlamudi has invented what I assess to be the first artificial intelligence architecture purpose-built to address this specific problem. He named and designed the DGP — Deterministic-Generative-Predictive — Clinical Revenue Architecture, a novel three-layer hybrid AI system for independent laboratory claim denial recovery.

The architecture operates as follows. The first layer is a deterministic symbolic policy engine encoding 847 compliance rules derived from Local Coverage Determinations (LCDs), National Coverage Determinations (NCDs), MolDX program requirements, DEX Z-code taxonomy, CPT codes 81400 through 81479 (the full genomic sequencing series), and Proprietary Laboratory Analysis (PLA) code documentation requirements. This layer produces a binary compliance determination and structured denial-risk classification for each claim, with each output traceable to a specific named policy rule — providing zero-hallucination policy interpretation and complete auditability.

The second layer applies a large language model for generative clinical reasoning. Critically, this LLM does not interpret coverage policy — that function has already been discharged, deterministically, by Layer 1. Instead, the generative layer reads available clinical documentation, retrieves evidence from 14 curated medical databases (including PubMed, ClinicalTrials.gov, the CMS LCD/NCD repository, and specialty pharmacogenomics references) in under 340 milliseconds via a parallel retrieval pipeline, and synthesizes this information into a compliant, payer-specific appeal brief in under 90 seconds. Every LLM output is logged with its input tokens, evidence provenance, and the Layer 1 policy citations that grounded it.

The third layer applies machine learning trained on historical EDI 835 remittance data to score outgoing claims for denial probability before submission. Claims exceeding a configurable risk threshold are routed for pre-submission correction, enabling proactive denial prevention rather than reactive appeals management. The system is fully integrated via FHIR R4, HL7 v2, and EDI 835/837 standards, deployed on Google Cloud Platform with a HIPAA Business Associate Agreement, and achieves native compliance with Texas Senate Bill 1188 and the Texas Responsible AI Governance Act (TRAIGA, HB 149) — two of the most comprehensive state-level AI governance laws currently in effect in the United States.

### **III. Why This Architecture Is Novel — Comparison to the Existing Field**

The novelty of the DGP architecture becomes clear when situated against the current state of the art. Prior published work on hybrid AI in healthcare — combining symbolic rule systems with neural or statistical components — addresses exclusively clinical diagnosis and treatment decision support: cholangitis management protocols, oncology clinical trial matching, medication recommendation systems, and IoMT-based patient monitoring. No prior published architecture applies this three-layer hybrid design to the healthcare administrative domain of claim denial management.

The commercial RCM AI market, though large (Waystar's 2024 IPO valued its AI-assisted RCM platform at \$3.5 billion; R1 RCM was acquired in 2024

for \$8.9 billion), is architected for hospital and health system billing, specifically diagnosis-related group (DRG) billing and physician evaluation-and-management (E/M) coding. These platforms do not encode MolDX program requirements, DEX Z-code stacking logic, or CPT 81400-through-81479-level medical necessity documentation requirements. When LLM-based RCM products attempt to interpret coverage policy without a deterministic rule layer, they hallucinate on specific LCD effective dates, MAC jurisdiction variations, and code-level coverage criteria — and a hallucinated policy citation in an appeal brief is sufficient to exhaust a laboratory's appeal rights on an otherwise valid claim.

Mr. Vadlamudi's deterministic-first design principle directly resolves this failure mode. The 847-rule engine ensures that the generative layer never makes a policy compliance judgment — it only generates clinical narrative grounded on compliance conclusions already resolved by Layer 1. This architectural decision is, to my assessment, both technically novel and practically essential for safe deployment in the healthcare billing context.

#### **IV. Why Mr. Vadlamudi's Continued Presence in the United States Is in the National Interest**

The problem Mr. Vadlamudi's architecture addresses is a national healthcare infrastructure problem, not a commercial product opportunity. The 2.76 times higher denial odds for independent versus hospital-based laboratories documented in the JAMA Network Open study represents a structural inequity in the US healthcare reimbursement system: the same NGS test, ordered for the same clinical indication, faces dramatically different reimbursement outcomes based solely on whether the performing laboratory is affiliated with a hospital system. AI-driven claim recovery that reduces this disparity has direct implications not only for laboratory economics but for community healthcare access.

Independent laboratories are disproportionately the sole diagnostic resource for rural communities served by Critical Access Hospitals, for addiction medicine practices relying on toxicology panels, and for specialty oncology practices performing NGS-guided treatment selection. The laboratory consolidation that the ACLA survey projects as a consequence of PAMA cuts combined with high denial rates would remove diagnostic capacity from precisely these communities. Mr. Vadlamudi's DGP architecture directly addresses the underlying economic problem driving that consolidation risk.

No existing commercially available AI system, and no system described in the published research literature, addresses the specific regulatory complexity of independent laboratory molecular diagnostic billing at the domain specificity and architectural rigor that Mr. Vadlamudi's platform provides. His continued presence and continued development of this technology in the United States — without the disruption of labor

certification barriers — directly benefits the patients, providers, and communities that depend on independent laboratory diagnostic services.

## V. Conclusion

Based on my independent expert assessment, I find that Rambabu Vadlamudi's DGP Clinical Revenue Architecture represents an extraordinary original contribution of major significance to the field of healthcare artificial intelligence. His work is technically distinguished, addressing a well-documented and quantified national problem that no prior published system solves. He brings a combination of deep healthcare IT operational experience — fifteen-plus years across St. Jude Children's Research Hospital, CIGNA, Teladoc Health, United Health Care, ECFMG, and Medifast — and genuine architectural innovation that is rare in any field. The national interest benefit of his continued work in the United States is, in my professional opinion, clear and substantial.

I strongly and without reservation recommend approval of Mr. Vadlamudi's EB-1A Extraordinary Ability petition.

Respectfully submitted,

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[LETTER WRITER NAME] [TITLE] [INSTITUTION / ORGANIZATION]  
[ADDRESS] [EMAIL] [PHONE] [DATE]

## LETTER 2 — CLINICAL LABORATORY DIRECTOR (PILOT PARTNER / POTENTIAL CUSTOMER)

**Best for:** Lab directors at HealthTrackRx, Principle Health Systems, or any independent CLIA-certified lab director in PGx/NGS billing **Strongest EB-1A criteria served:** C5 (original contributions from the end-user perspective), C8 (leading role — corroborates Ardia as a distinguished organization serving a real market) **Note:** This letter is most powerful after the HealthTrackRx pilot launches (Q2 2026). The [PILOT RESULTS] paragraph should be filled in with actual outcomes data at that time.

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[DATE]

[LETTER WRITER NAME] [TITLE / POSITION] [LABORATORY NAME]  
[ADDRESS LINE 1] [CITY, STATE ZIP] [CLIA NUMBER: XX-DXXXXXX]  
[EMAIL] [PHONE]

U.S. Citizenship and Immigration Services National Benefits Center  
[Service Center Address]

## **RE: Expert Letter in Support of EB-1A Extraordinary Ability Petition for Rambabu Vadlamudi**

Dear USCIS Officer:

I am writing to offer my professional assessment in support of the EB-1A Extraordinary Ability petition filed on behalf of Rambabu Vadlamudi, Founder and Enterprise Architect of Ardia Health Labs. My name is [NAME], and I serve as [Laboratory Director / Chief Scientific Officer / Director of Revenue Cycle Operations] at [LABORATORY NAME], a CLIA-certified independent clinical laboratory located in [CITY, STATE], CLIA number [XXXXXXXX]. Our laboratory processes approximately [X] pharmacogenomics, next-generation sequencing, and molecular diagnostic claims monthly and generates approximately [\$X million / \$X thousand] in annual laboratory revenue from this testing category. I have [X] years of experience in laboratory operations, [X] years in clinical billing and revenue cycle management, and [relevant credentials, board certifications, or professional affiliations]. I offer this letter based on my direct professional knowledge of the problem Mr. Vadlamudi's work addresses and my assessment of its significance to laboratories like mine.

### **I. The Reimbursement Crisis Facing Independent Laboratories**

I can speak from direct operational experience about the billing crisis Mr. Vadlamudi has dedicated his work to solving. Independent clinical laboratories — institutions like mine, operating outside of hospital systems — face a structural disadvantage in molecular diagnostic claim reimbursement that has materially threatened our financial viability and our ability to serve patients.

The numbers published in the research literature match what I observe in our own operations. A 2024 XiFin analysis of more than 20 million laboratory claims documented a molecular diagnostics denial rate of 35.3% — the highest of any healthcare specialty. A 2025 Georgetown University study published in JAMA Network Open confirmed, after controlling for payer mix, test complexity, and patient demographics, that independent laboratories face 2.76 times higher odds of claim denial than hospital-based laboratories. [PERSONALIZE: Insert one to two sentences describing your laboratory's specific denial experience, e.g.: "At [LABORATORY NAME], our pharmacogenomics denial rate has run as high as [X]% in [time period], representing approximately \$[X] in monthly preventable revenue loss."]

The reason most of this revenue is never recovered is not that the claims lack clinical merit — it is that mounting a proper appeal requires identifying the exact Local Coverage Determination, MolDX policy, or DEX Z-code requirement applicable to the denied claim, obtaining physician attestation, and drafting a medical necessity narrative that cites the correct policy version. For a billing department of [X] FTE, handling [X] denial letters per

month, this is simply not feasible. The Healthcare Financial Management Association has documented that 65% of denied laboratory claims are never appealed, despite appeal win rates of 50% to 80.7%. That gap represents money our laboratory has written off not because the claims were wrong, but because we lacked the specialized resources to fight them efficiently.

## **II. Why Mr. Vadlamudi's DGP Architecture Is an Original and Significant Contribution**

Mr. Vadlamudi invented the DGP — Deterministic-Generative-Predictive — Clinical Revenue Architecture, and in my professional assessment as a laboratory operator, this is the first platform I have encountered that is genuinely architected for our billing environment.

The architecture has three layers. The first is a deterministic policy engine encoding 847 compliance rules covering all LCD and NCD policies, MolDX program requirements, DEX Z-codes, CPT codes 81400 through 81479 for genomic sequencing, and PLA code documentation requirements. This is the layer that matters most to me: the MolDX program alone involves more than 75,000 molecular diagnostic tests mapped to under 200 CPT codes, with an estimated 35-plus percent coding error rate. No general-purpose billing AI product I have evaluated has a working model of this regulatory framework. Mr. Vadlamudi's architecture encodes it deterministically, rule by rule, with each compliance determination traceable to a named policy document.

The second layer uses a large language model to generate appeal briefs — but only after the policy determination has been resolved by Layer 1. This design decision is critical: the LLM generates the clinical narrative of the appeal; it does not decide whether the claim should be covered. The platform pulls evidence from 14 medical databases in under 340 milliseconds and generates a complete, payer-formatted appeal brief in under 90 seconds. The third layer scores claims before submission using machine learning trained on historical remittance data, identifying high-denial-risk claims for correction before they are filed.

In [X] years of laboratory operations, I have not encountered any existing platform — commercial or open-source — that combines these three capabilities specifically for independent laboratory PGx and NGS billing. The large commercial RCM vendors (Waystar, R1 RCM) are built for hospital DRG billing; they do not know what a DEX Z-code is, let alone encode coverage policy for CPT 81479. Mr. Vadlamudi's architecture was designed specifically for our billing environment, and it addresses a technical and regulatory gap that has cost the independent laboratory industry an estimated \$10 to \$12 billion annually.

Following the deployment of Mr. Vadlamudi's Ardia Health platform at [LABORATORY NAME] in [MONTH YEAR], we observed the following outcomes over a [X]-month period: [INSERT SPECIFIC METRICS, e.g., "Our

molecular diagnostics denial rate decreased from [X]% to [Y]%, representing a \$[Z] monthly improvement in recovered revenue. Our average appeal preparation time decreased from [X] hours to under [Y] minutes per claim. We successfully appealed [X] previously written-off denials, recovering \$[Z] in revenue. The platform identified [X] high-denial-risk claims before submission, enabling pre-submission correction and avoiding [X] anticipated denials.”] These results demonstrate that Mr. Vadlamudi’s architecture delivers the practical impact its design intends, in the real operating environment of an independent clinical laboratory.

### **III. The National Importance of Mr. Vadlamudi’s Work**

This is not a niche problem affecting a small market segment. There are more than 7,000 CLIA-certified independent laboratories in the United States. According to the American Clinical Laboratory Association’s 2024 member survey, more than 50% of independent laboratory directors reported considering consolidation or sale if the PAMA 2027 reimbursement cuts take effect as projected — cumulative cuts of up to 45% by 2029. When combined with the existing 35.3% denial rate in molecular diagnostics, the financial pressure on independent laboratories is existential for many facilities.

What consolidation or closure means in practice is this: in the communities we serve, an independent laboratory closure is not replaced by a hospital system laboratory. Rural communities, addiction medicine clinics, and specialty oncology practices lose access to rapid, proximate molecular diagnostic testing. The result is delayed pharmacogenomics-guided prescribing, disrupted oncology treatment monitoring, and reduced access to precision diagnostics for the patients who most depend on independent laboratory services.

Mr. Vadlamudi’s DGP architecture directly addresses the denial-rate side of this financial equation. By automating the regulatory compliance and appeal brief generation that independent laboratories currently lack the resources to perform at scale, his platform enables laboratories to recover revenue that is rightfully owed — and to continue operating the diagnostic services their communities depend on. This is not a commercial convenience; it is a healthcare infrastructure necessity.

### **IV. Conclusion**

Rambabu Vadlamudi is a rare combination: a practitioner with deep operational experience in the healthcare IT systems that govern our billing environment — having served in healthcare technology roles at St. Jude Children’s Research Hospital, CIGNA, Teladoc Health, United Health Care, ECFMG, and Medifast — and an original technologist who has built a platform that no prior developer thought to build. The problem he is solving



is real, the quantification of that problem is robust and peer-reviewed, and the architecture he has invented directly addresses a gap that costs the US independent laboratory sector billions of dollars annually.

Independent laboratories serve communities and patients that hospital-based diagnostics do not reach. Mr. Vadlamudi's work is directly in service of those communities. The United States healthcare system benefits substantially from his continued presence here, unimpeded by labor certification barriers, to complete and deploy this work.

I strongly recommend approval of his EB-1A Extraordinary Ability petition without reservation.

Respectfully submitted,

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[LETTER WRITER NAME] [TITLE] [LABORATORY NAME] [CLIA NUMBER]  
[ADDRESS] [EMAIL] [PHONE] [DATE]

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## Usage Notes

### For Both Letters

1. Fill in all bracketed fields before sending to the letter writer.
2. Print on institutional letterhead.
3. The letter writer should sign and date in original ink (wet signature) for USCIS submission.
4. Include the letter writer's CV or abbreviated bio as an exhibit tab in the petition package.

### For Letter 1 (Researcher/Expert)

- The strongest version of this letter comes from someone who has published on the exact problem Ram's work addresses — ideally the Georgetown University PI of the JAMA 2025 NGS denial study, or a XiFin principal analyst.
- The paragraph "I am independent of Mr. Vadlamudi" is important for USCIS credibility — preserve it or adapt it truthfully.
- If the letter writer has published on hybrid AI in healthcare, add a sentence in Section III citing their own work and explaining how Ram's contribution differs from or extends it.

### For Letter 2 (Lab Director)

- This letter becomes dramatically more powerful after the HealthTrackRx pilot produces outcome data. The [PILOT RESULTS PARAGRAPH] should be filled in with real numbers.

- The letter writer's CLIA number and laboratory scale (monthly claim volume, annual revenue) establish that this is a peer professional, not a casual contact. Include them.
- If the letter writer is the HealthTrackRx lab director, reference the specific pilot engagement and timeline.

### **Petition Package Placement**

- These letters support Criterion 5 (Original Contributions of Major Significance) primarily.
- They also reinforce Criterion 8 (Leading Role) by corroborating Ardia Health Labs as a real, operating organization addressing a significant market.
- Tab each letter separately in the petition exhibit index: "Exhibit [X] — Expert Letter of [NAME], [TITLE], [INSTITUTION]."